

GaN Based Synchronous Buck Converter with an Improved Gate Driver

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Review & Date:

Review-II · 30.09.2026 (Wednesday)

Guide's Signature with Date

Signature: _____

Date: _____

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Problem Statement & Background

Problem Statement:

When one GaN transistor switches, its fast voltage swing pushes charge into the gate of the other transistor, the one that is supposed to stay off. That gate rises to **1.65 V, and only 1.4 V is needed to turn it on.** Both devices conduct at once and the supply shorts through them. (That 1.65 V is our device model's; how much the result depends on the model is itself a finding, and it has its own slide.)

Background & Significance:

GaN half-bridges sit inside EV inverters and battery chargers. GaN is chosen because it switches fast, and that speed is exactly what causes this fault.

Existing Solutions:

Gate drivers that turn the device on in steps, clamp the off gate down, adjust the gap between the two switching edges, and hold the gate at -2 V [1]–[4]. Reported: 30.5 % less overshoot, 75 % less turn-off loss [5]–[7].

Limitations / Research Gap:

The reported gains do not separate what the re-tuning is worth. Those gains mix two different things: picking one good setting, and changing the setting while the converter runs. Only the second needs a sensor, a lookup table and a controller.

Aim, and how we approached it

Aim:

To find out how much of a gate driver's benefit needs the driver to re-tune itself while the converter is running, and how much comes from choosing one good setting and leaving it fixed.

Proposed Solution:

A GaN buck converter driven by a **segmented gate driver**: 8 pull-up steps, 8 pull-down steps, an adjustable dead time, a Miller clamp and a -2 V off rail. Every one of them is a setting we can change and measure.

How we approached it:

- 1. Build the converter and check it converts.** 100 V DC in, 48.50 V DC out at 4.869 A: 236.26 W delivered, 97.49 % efficient.
- 2. Recreate the fault.** At the fastest setting the gate that should be off reaches 1.65 V, against a 1.4 V turn-on threshold.
- 3. Fix it one change at a time,** measuring each change on its own run, not all at once.
- 4. Ask whether one fixed setting is enough,** by running the driver at two operating points and seeing whether the best setting moves.

Scope and tools:

In scope: the converter, the driver, its settings, and the FPGA controller. **Tools:** ngspice for every simulation, LTspice to draw the circuit and re-check, Vivado for the FPGA.

System Architecture

The driver is drawn as `kicad/gan_segdrv.kicad_sch` and placed twice in `kicad/gan_buck.kicad_sch` — once per gate.

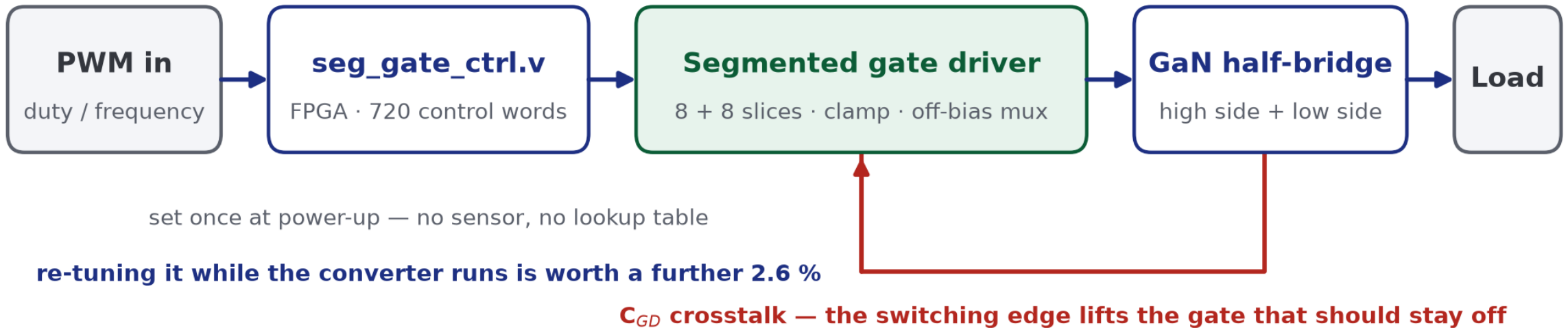


Fig. 1 — Drawn diagram, not simulation output. One signal path: a PWM command, the FPGA controller that holds the control word, the segmented driver that shapes the edge, and the half-bridge it drives. The driver in green is what this project designs; it is drawn once as [kicad/gan_segdrv.kicad_sch](#) and placed twice in [kicad/gan_buck.kicad_sch](#), one per gate. The control word is set at power-up and never changed — re-tuning it while the converter runs is the 2.6 % this project measures. In red, the fault: the switching edge couples through C_{GD} into the gate that should stay off.

The circuit we simulate

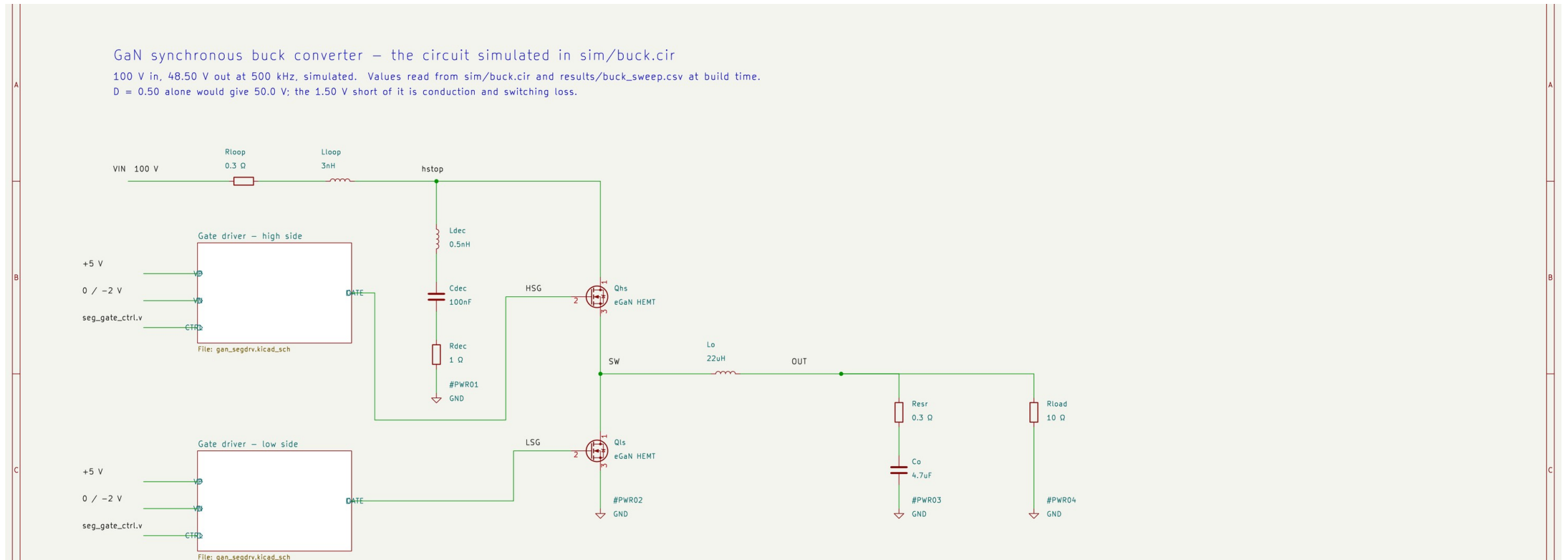


Fig. 2 — Drawn diagram, not simulation output. GaN synchronous buck converter, drawn from `sim/buck.cir` by [scripts/kicad_schematic.py](https://github.com/keithmcbryde/scripts/kicad_schematic.py) into [kicad/gan_buck.kicad_sch](https://github.com/keithmcbryde/scripts/kicad_gan_buck.kicad_sch) — every value on it is read out of the netlist at build time. “Buck” means step-down: 100 V in, 48.50 V out. Left to right: the 100 V supply with its power-loop R and L, the damped bus decoupling branch (Ldec, Cdec, Rdec at 1 Ω), the two GaN HEMTs whose gates the segmented drivers drive, the output filter and a 10 Ω load.

What we add that they do not have

Same command, same segmented output stage, same power stage. Three blocks added — in green.

	BASE PAPER — Zhang et al., ISPSD 2020	OURS
Control of the driver	one analogue bias resistor, set at fabrication	seg_gate_ctrl.v — digital, re-writable while it runs
Output stage	7 + 7 slices, two stages	8 + 8 slices, thermometer coded
Active Miller clamp	none	always on
Gate off rail	0 V	selectable 0 V / -2 V
Crosstalk margin at 100 V / 10 A	+0.407 V	+2.576 V

Shaded: the three blocks we add. Everything else is theirs, unchanged — same command, same power stage. 6.3× the margin, from one fixed setting.

Fig. 3 — Drawn diagram, not simulation output. The five things that differ, and nothing else. Their column is the base paper as we rebuilt it in [models/zhangdrv.lib](#); ours is [models/segdrv.lib](#). Shaded: the three we add — digital control in place of one fixed resistor, an always-on Miller clamp, and an off rail selectable to -2 V. The command, the power stage and the idea of segmenting the output are all theirs. The last row is what those three buy at the headline corner, read from [results/headtohead.csv](#): +0.407 V becomes +2.576 V, from one fixed setting.

What differs in the circuit, theirs and ours

The converter on top; underneath it, what sits behind its two gates in their design and in ours. Both additions are established practice [1]-[4]; the base paper uses neither. What is measured here is what each is worth on this converter.

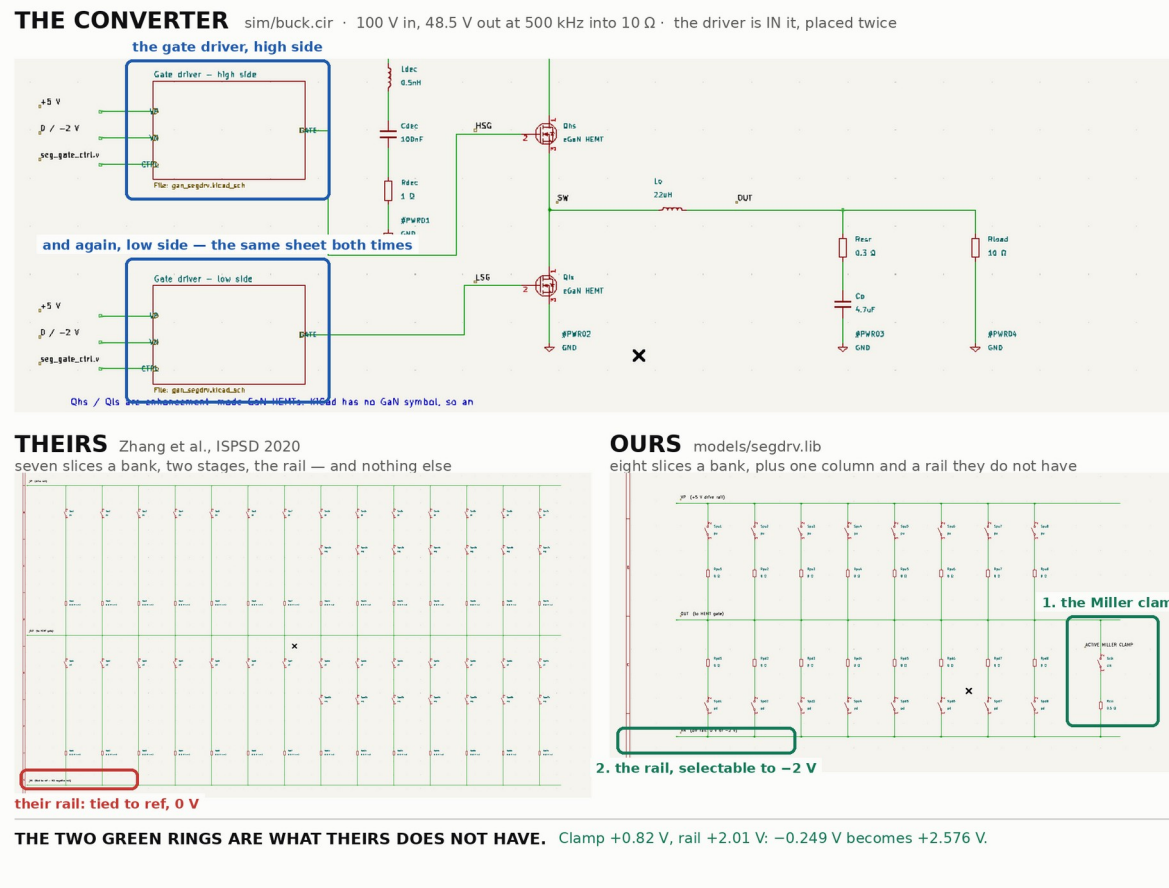


Fig. 4 — Top: [kicad/gan_buck.kicad_sch](#), with the two gates of the half-bridge ringed — HSG and LSG are what a gate driver drives, and everything below is what sits behind them. Bottom left, theirs ([kicad/gan_zhangdrv.kicad_sch](#)): seven slices a bank in two stages, and a rail their own sheet labels “tied to ref — NO negative rail”. Bottom right, ours ([kicad/gan_segdrv.kicad_sch](#)): the same eight-slice output stage, plus the active Miller clamp and an off rail selectable to -2 V. The clamp is worth $+0.82$ V of crosstalk margin and the rail $+2.01$ V; together -0.249 V becomes $+2.576$ V. Annotated screen captures of KiCad 7.0.11, not redrawings, and not a size comparison — the three sheets open at different zoom.

How every number on these slides was made

Every comparison is the same file with one thing changed. The following is held identical; the listed parameter is the only one that varies.

HELD IDENTICAL

the netlist [sim/dpt.cir](#), byte for byte — not a copy, not a variant

the device [models/egan.lib](#) on both sides, same V_{th} , same $R_{es(on)}$, same C_{9e}

the parasitics 3 nH power loop, 0.3 Ω loop resistance, same gate inductance

the operating point 100 V bus, 10 A load, 25 °C junction

the solver same timestep, same options, same initial conditions

WHERE IT RAN ngspice-42 and KiCad 7.0.11 on a Debian container, headless, no hardware in the loop. Both are open source; there is no licensed tool and no vendor model anywhere in this project.

THE ONE THING THAT MOVES

theirs vs ours the driver subcircuit, and nothing else:
[models/zhangdrv.lib](#) ↔ [models/segdrv.lib](#)

GaN vs silicon the device model only, $R_{es(on)}$ matched at the 25 m Ω class

clamp on/off one parameter, CLKEN; off rail, one parameter, VNEG

Fig. 6 — Their driver is additionally given a freedom its own paper does not have: [scripts/headtohead.py](#) searches both of its controls at every corner and it is run at the setting that search finds optimal for it, read out of [results/headtohead.txt](#). Ours runs one fixed control word at every corner. Any remaining difference is the driver. [results/RESULTS-SUMMARY.txt](#) names the script behind each number, and [review/check_consistency.py](#) fails the build if a slide and the data disagree.

The base paper, simulated — their driver, their result

[models/zhangdrv.lib](#) on [sim/dpt.cir](#). Seven slices per bank in two stages, no clamp branch, no negative off rail.

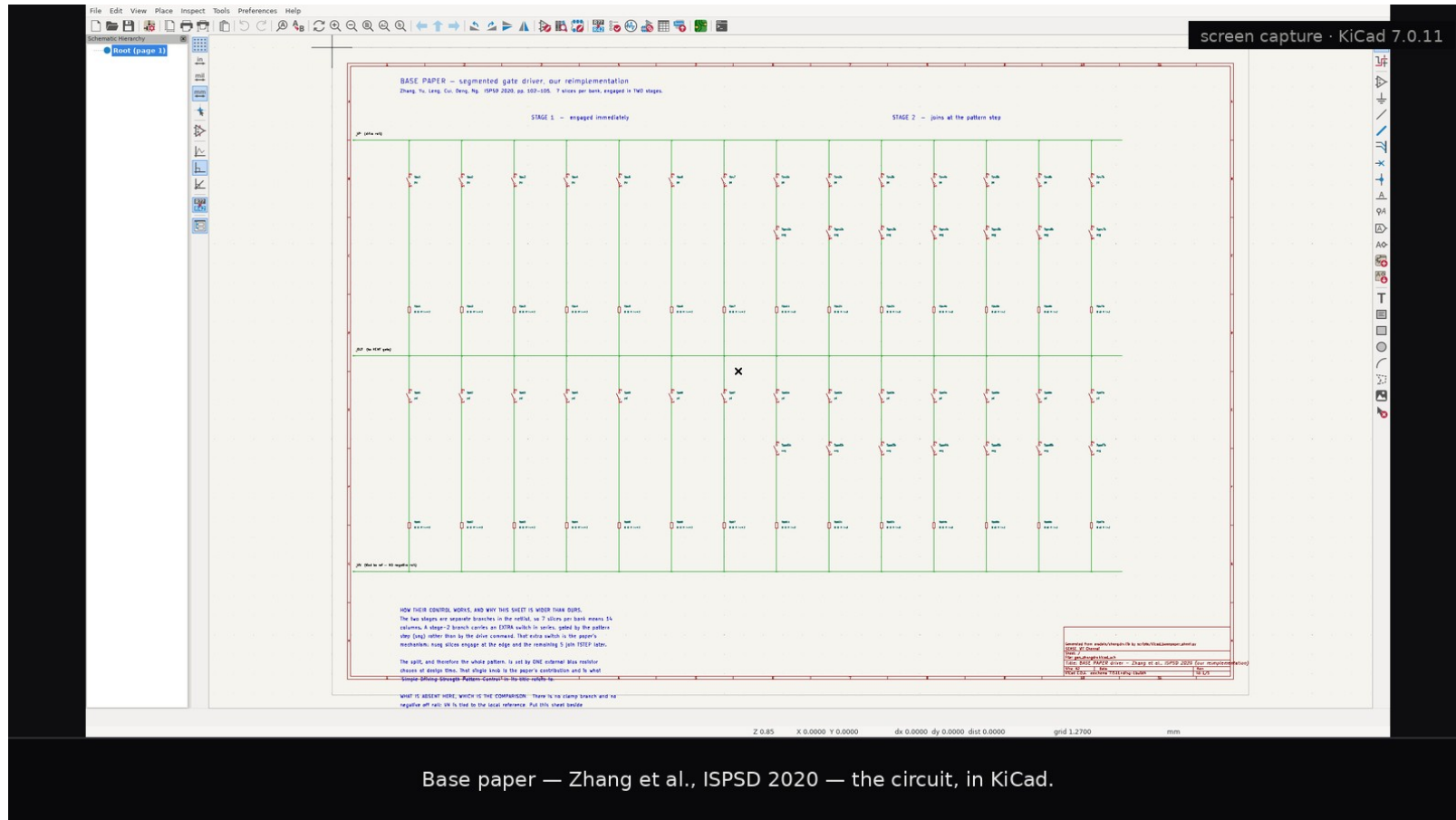
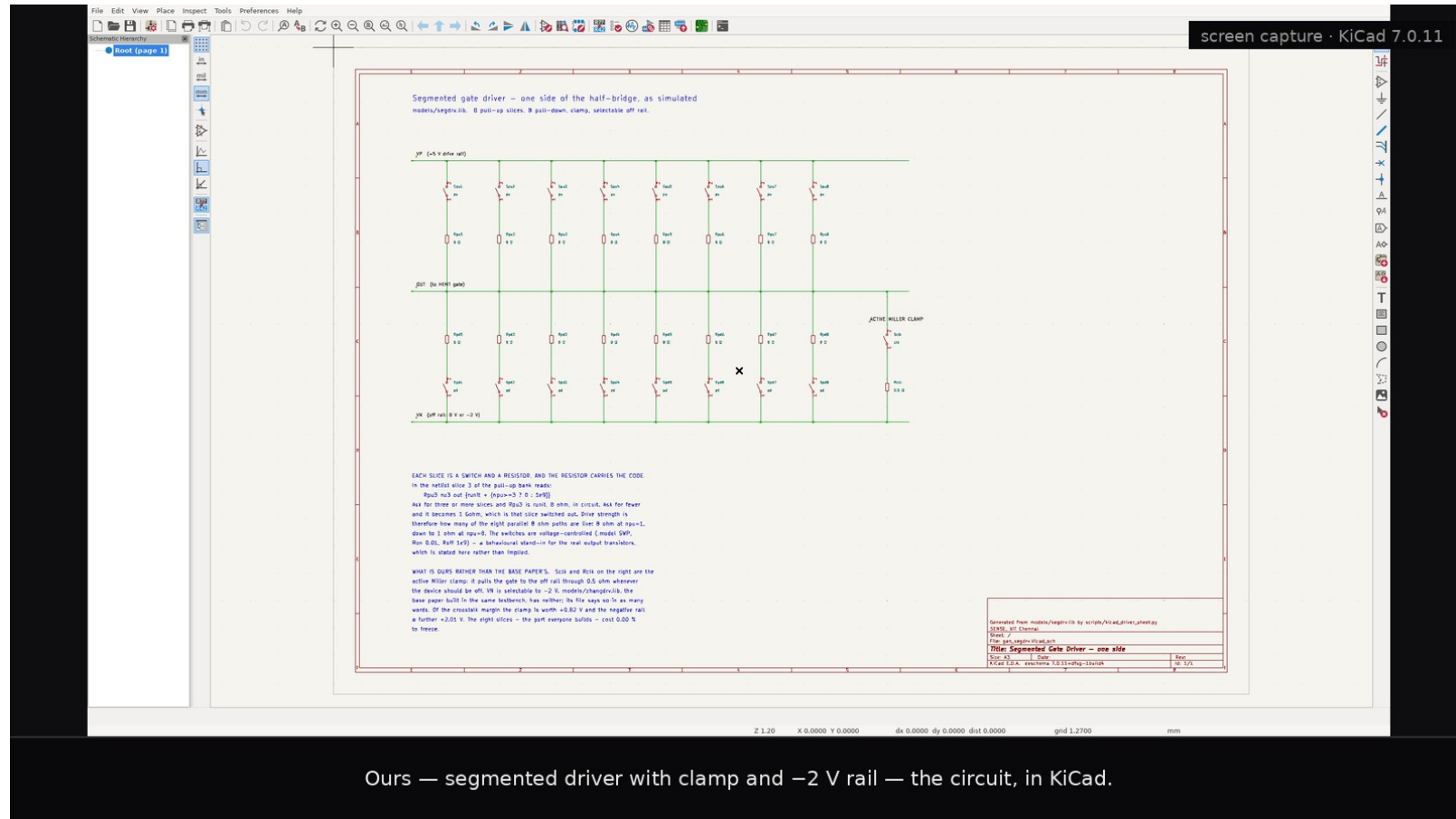


Fig. 7 — Their circuit in KiCad ([kicad/gan_zhangdrv.kicad_sch](#)), ngspice running it, and the waveform it produced. The OFF device's gate peaks at +0.993 V against a 1.400 V threshold — 0.407 V of margin. Run at nseg=2, tstep=4n, the setting [scripts/headtohead.py](#) found optimal for their driver at this corner, read from [results/headtohead.txt](#): their own paper sets one bias resistor once at design time and gets no per-corner choice at all. 35 s.

Ours, simulated — same bench, same corner, same axes

[models/segdrv.lib](#) on the same [sim/dpt.cir](#). Eight slices per bank, active Miller clamp, -2 V off rail.



Ours — segmented driver with clamp and -2 V rail — the circuit, in KiCad.

Fig. 8 — The same sequence over our driver. The gate peaks at -1.176 V — 2.576 V of margin, $6.3\times$ theirs. One fixed control word, the same one used at every corner in the study, not re-tuned for this run. The plot axes are identical to the previous slide's and fixed in the script, so the two are directly comparable. 35 s.

Demo — the driver, built and measured

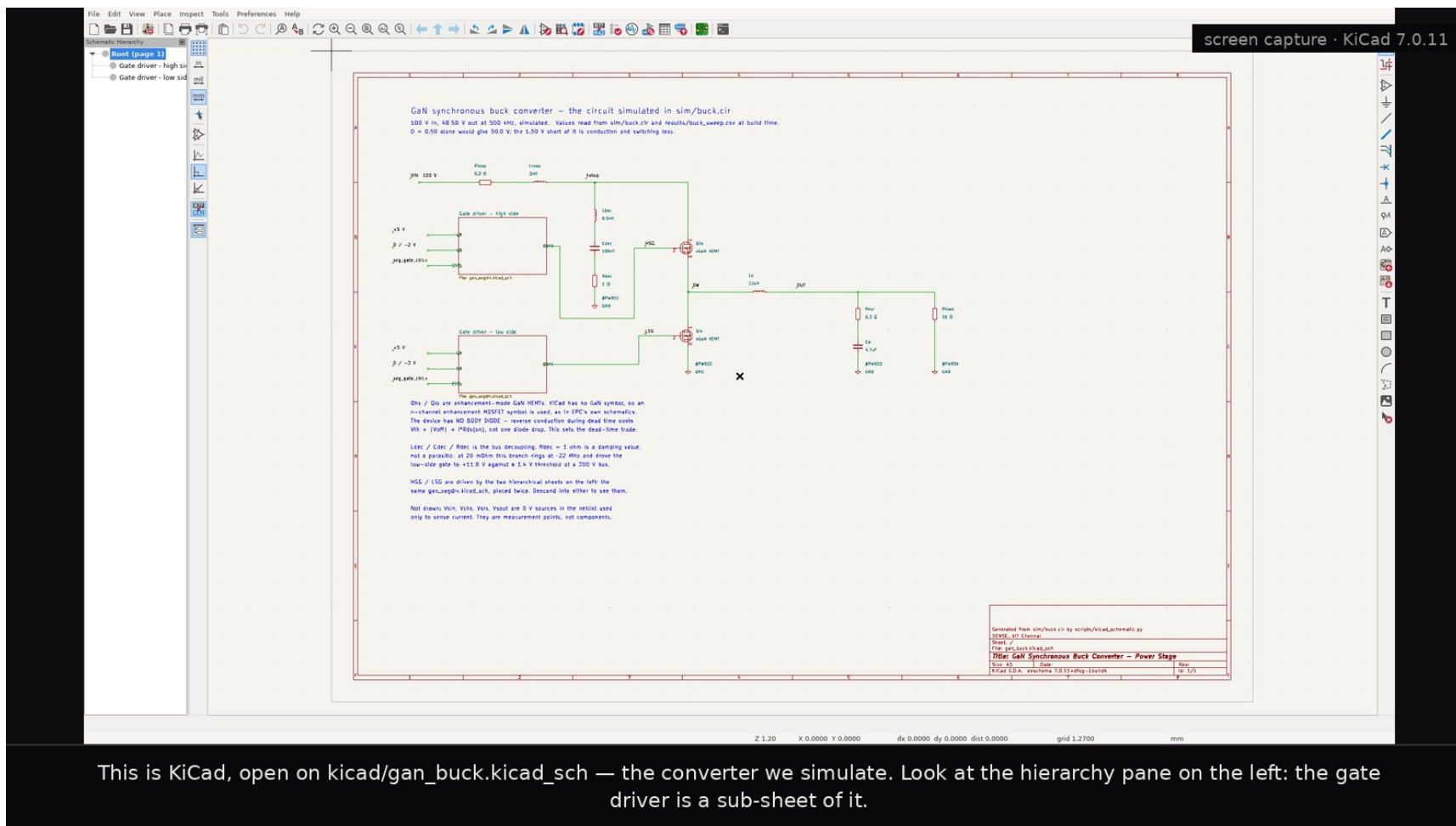


Fig. 9 — Four sections, 100 s. The circuit as its KiCad sheet ([kicad/gan_buck.kicad_sch](#)), the tools named and versioned, the output with the threshold, the peak and the margin annotated, and the base paper's driver run against ours on the same bench at 100 V / 10 A / 25 °C. Theirs peaks +0.993 V, 0.407 V short of the 1.400 V threshold; ours peaks −1.176 V, 2.576 V clear. The converter it drives: 48.50 V out at 236.26 W, 97.49 % efficient. Three ngspice runs made while the film was building — rebuilt by [scripts/demo_review2.py](#), so every number on screen is computed, not captioned.

The same simulation, run live

bash [proof/LIVE-SIM.sh](#) — two ngspice transients, about ten seconds, on the presentation machine.

```
$ bash proof/LIVE-SIM.sh 5.0 s

GaN half-bridge -- crosstalk, simulated live
-----
machine  vm      2026-09-27 13:24:17
spice    ** ngspice-42 : Circuit level simulation program
circuit  sim/dpt.cir      device  models/egan.lib
changing ONLY  CLKEN (clamp) and VNEG (off rail)

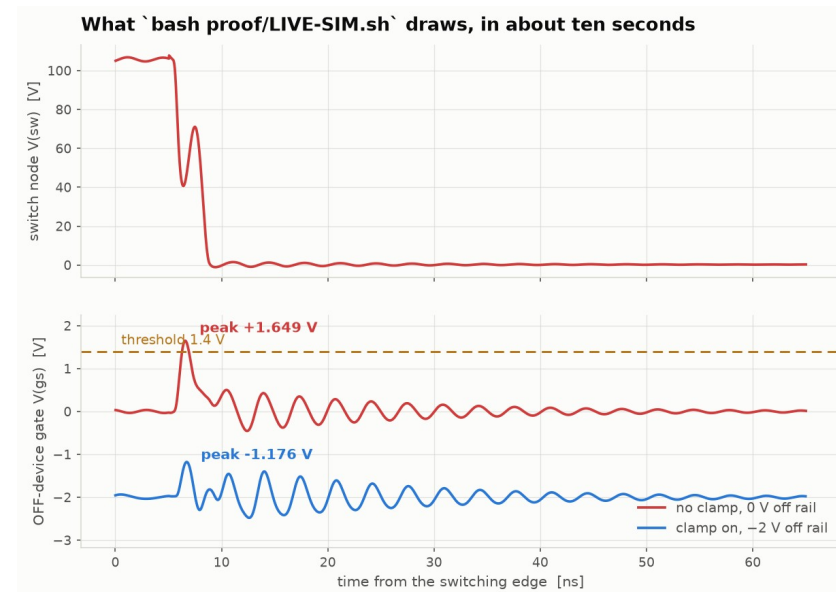
(a) fastest drive, no clamp, 0 V off rail ...
    OFF-device gate reaches +1.6488 V  against a 1.400 V threshold
    margin -0.2488 V    false_turn_on = 1  SHOOT-THROUGH

(b) same circuit, clamp on, -2 V off rail ...
    OFF-device gate reaches -1.1759 V
    margin +2.5759 V    false_turn_on = 0  SAFE

-----
                                on the slide    this run
OFF gate, no clamp [V]          1.65           1.6488
margin, shipped [V]             2.576          2.5759
-----
two ngspice transients, 3.7 s wall clock on this machine

drawing the waveforms those two runs produced ...
```

recorded run of the same command



the waveforms it produces

Fig. 10 — Simulated in ngspice. Two runs of one circuit file. Between them only CLKEN (the Miller clamp) and VNEG (the gate off rail) change, so no other parameter varies. It prints what it measured beside what these slides claim — OFF gate 1.649 V against the deck's 1.65, margin 2.576 V against the deck's 2.576 — then draws the waveforms those two runs produced. 3.8 s of that is ngspice; the rest is drawing. [scripts/live_demo.py](#).

Crosstalk simulation — the fault, and the fix

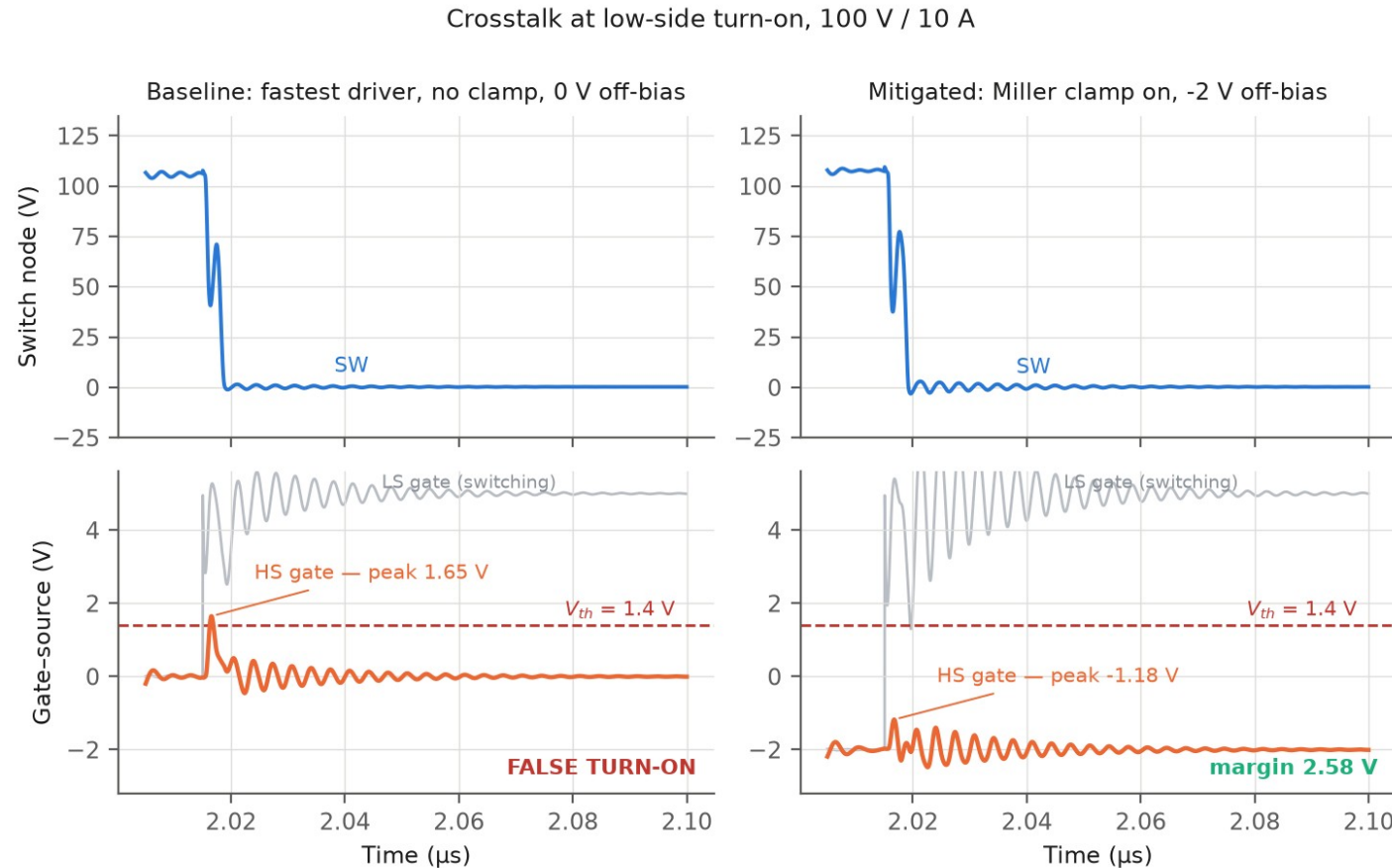


Fig. 11 — Simulated in ngspice. Two runs of [sim/dpt.cir](#). Top row: the switch node falls from 106 V in 2.90 ns, peaking at 124 V/ns — that dv/dt drives $i = C_{GD} \cdot dv/dt$ into the OFF device's gate, and is the cause. Bottom row: that gate. Left, without the clamp, it rests at +0.005 V, is lifted +1.643 V and peaks at +1.649 V, past the 1.4 V threshold — `false_turn_on = 1`. Right, with the clamp and the -2 V rail, it rests at -1.941 V, is lifted only +0.765 V and peaks at -1.176 V — `false_turn_on = 0`. The two fixes are not one mechanism: the rail moves where the gate starts, the clamp shortens the lift by giving the injected charge a 0.5 Ω path out. [scripts/waveform_anatomy.py](#).

Silicon, the base paper, and ours — six parameters

Column 1 → 2 changes only the device. Column 2 → 3 changes only the control.

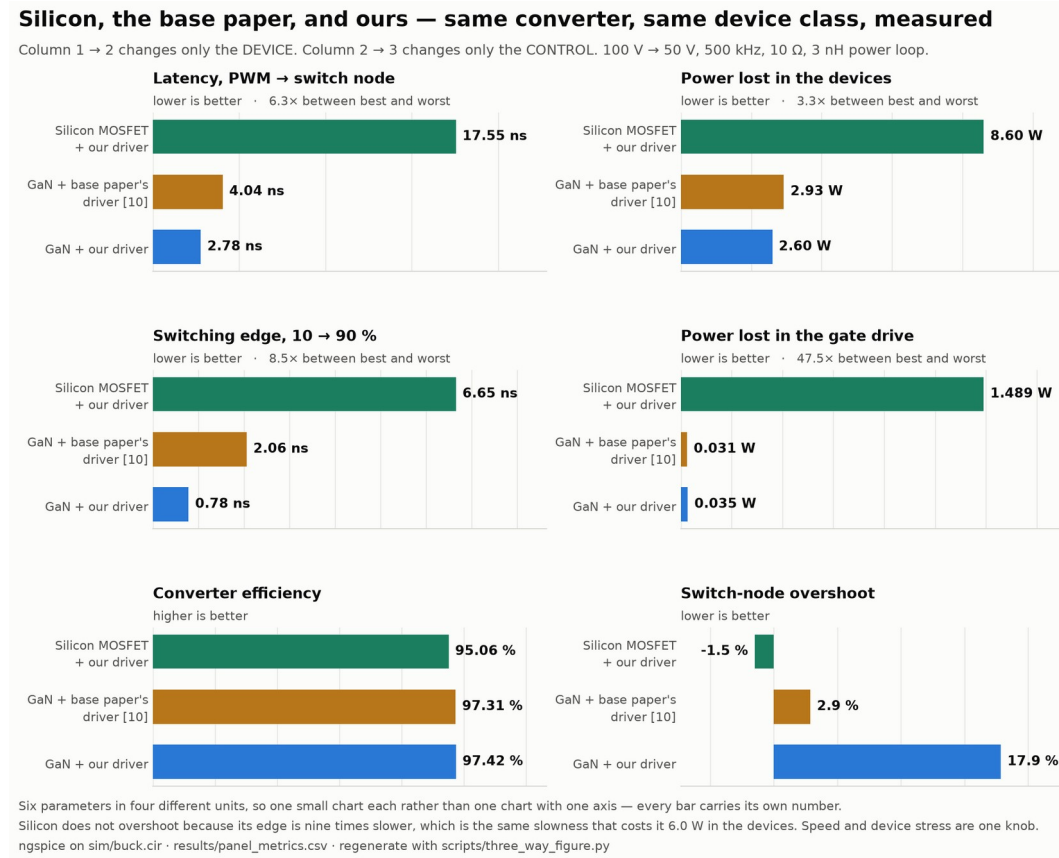


Fig. 12 — Simulated in ngspice. Latency 17.55 → 4.04 → 2.78 ns and device power 8.60 → 2.93 → 2.60 W, left to right. Against silicon that is 6.3× and 3.3×; against the base paper's control on the same GaN device, 1.5× and 1.1×. The device swap buys most of it; the control swap buys the rest. Overshoot is the row that runs the other way, and it runs that way for the same reason the other five do not: silicon does not overshoot because its edge is nine times slower, and that slowness is what costs it 6.0 W. Speed and device stress are one knob.
[scripts/three_way_figure.py](#), from [results/panel_metrics.csv](#).

Head to head with the base paper

5.5× to 12.4×, and the lead widens as the corner gets harder.

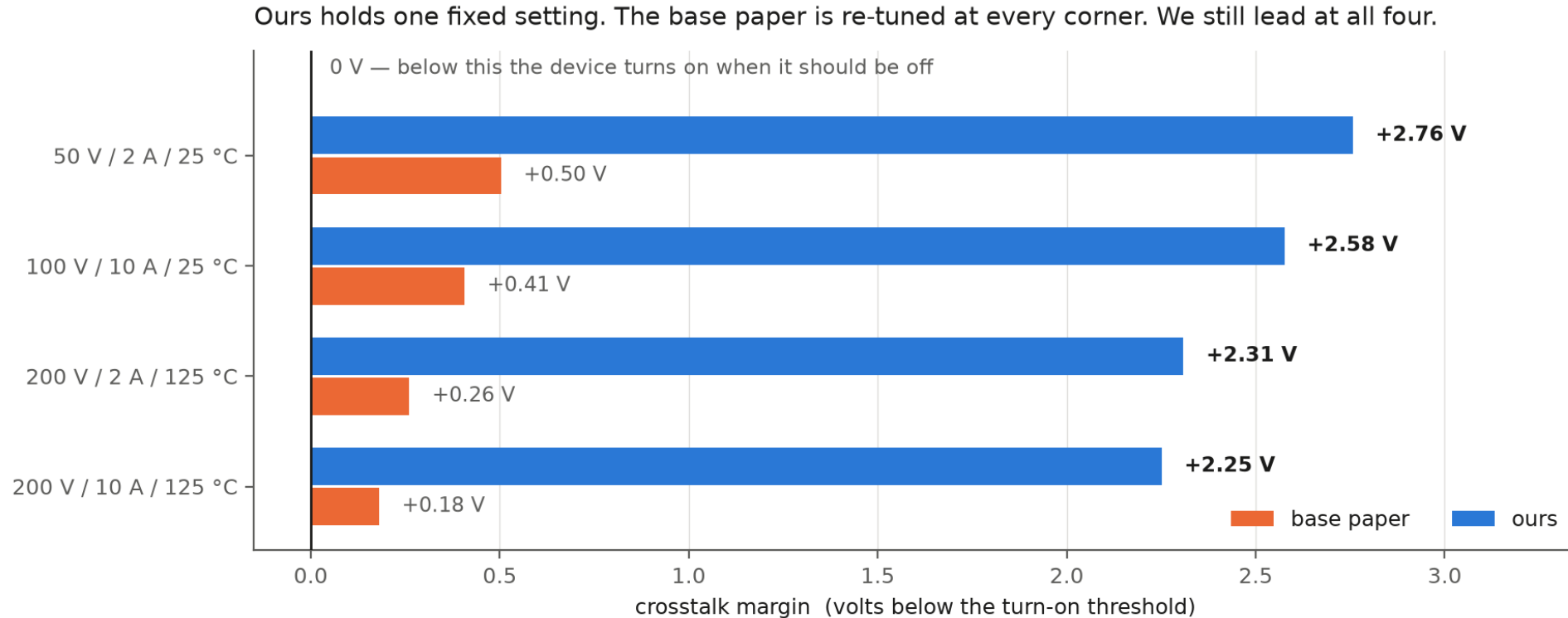


Fig. 13 — Simulated in ngspice. [scripts/headtohead.py](#): same deck, same GaN, same parasitics; only the driver is swapped. We hold one fixed control word at all four corners; they are re-optimised at every corner, which is more freedom than their own design has. Their margin falls +0.50 → +0.18 V from the mildest corner to the hottest and ours barely moves, because the clamp's action is independent of how hot the device is. Our switch node also slews about twice as fast, so the margin is not bought with switching speed. One caveat: [models/zhangdrv.lib](#) is our implementation of the scheme as they (their netlist is not published), run in our testbench with our parasitics. It is not 5.5–12.4× their measured result, and we do not claim it is.

Work Completed — 90 %

13 blocks, weighted by how much of the project each is. 11 are finished. The weights are stated so the total can be checked.

	Block of work	Weight	Evidence, or why not yet
✓	The converter itself, built and converting	6	100 V → 48.50 V at 4.869 A, 97.49 % efficient (buck.cir)
✓	The device choice justified against silicon	8	6.2 W vs 12.6 W at 500 kHz, on the shipped word; 3 sweeps, 4 duty profiles
✓	The base paper implemented, not just cited	8	zhangdrv.lib in our deck and in the converter — at 100 V only
✓	The segmented driver; the fault reproduced and fixed	12	−0.249 V → +2.576 V, one change at a time
✓	The full study: 720 words × 4 corners	8	66,924 transients over 36 corners — all on dpt.cir, not the converter
✓	How much controller that justifies	8	ladder + leave-one-corner-out; two implementations agree
✓	FPGA: RTL written, verified, synthesised, timing met	12	8 properties; 20 LUT / 20 FF; 200 MHz, 1.996 ns slack
✓	The two halves made to meet in one simulation	5	RTL VCD drives the SPICE slices; agree to 0.081 V
✓	Closed-loop regulation, disturbed on purpose	6	type-III loop: 0.02 % error through a 2x load and a 20 % line step
✓	Transistor-level output stage, on a real PDK	7	SKY130 5 V devices: sign and ordering of the result both survive
—	Place-and-route on a chosen board	3	Needs real package pins and an MMCM for the clock
✓	The converter swept across its stated envelope	10	8 bus×load points: margin +2.14 to +2.61 V everywhere, and the envelope corrected to 50–150 V because a 200 V part cannot run a 200 V bus
—	A hardware half-bridge, measured	7	Review-III. Until then this is a simulation study, and is titled as one

The simulation study is complete. What remains needs a bench: place-and-route on a chosen board, and one measured half-bridge. Neither can be obtained by further simulation.

90 of 100 by the count above. The weights are stated so the total can be recomputed. Score the hardware higher than 7 and the number falls — that is a fair reading, and it is the one we cannot answer without a bench.

Nothing has been measured on silicon.

References (1-15)

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The papers this design engages with directly. [10] is the base paper, reimplemented in [models/zhangdrv.lib](#) and measured against throughout.

Thank you

GaN Based Synchronous Buck Converter with an Improved Gate Driver

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Guide: Dr. Bindu — SENSE, VIT Chennai

Everything in this deck runs on request: github.com/Amritha902/gan-driver